



AROR UNIVERSITY
OF ART, ARCHITECTURE,
DESIGN & HERITAGE,
SUKKUR, SINDH

Department of Artificial Intelligence & Multimedia Gaming
CSC-207: Database Systems

Lab # 10: 1. Database Backup & Recovery
2. Generating Schema via ER-Diagram

Objectives

- Perform a full database backup using MySQL Workbench.
- Restore a database from a backup file.
- Design an **Entity-Relationship (ER) Diagram** for a given problem statement.
- Convert the ER diagram into a **MySQL schema** with proper tables, keys, and relationships.
- Implement the schema in **MySQL Workbench**.

Lab Tasks for Backup and Recovery

Task 1: Performing a Full Database Backup (Export)

1. **Open MySQL Workbench** and connect to your MySQL server.
2. **Navigate to the "Administration"** tab in the bottom-left panel.
3. **Click on "Data Export"** under the "Management" section.
4. **Select the database(s)** you want to back up from the list.
5. **Choose export options:**
 - **Export to Self-Contained File:** Saves the entire database as a single .sql file.
 - **Dump Structure and Data:** Ensures both schema and data are included.
 - (Optional) **Export to a project folder** for better organization.
 -

6. Click **"Start Export"** and wait for the process to complete.
 7. **Verify the backup file** in the specified location.
-

Task 2: Restoring a Database from Backup (Import)

1. **Open MySQL Workbench** and connect to your MySQL server.
 2. **Navigate to the "Administration"** tab.
 3. **Click on "Data Import/Restore"** under the "Management" section.
 4. **Select "Import from Self-Contained File"** and browse to your backup file (.sql).
 5. **Choose the target database:**
 - **New Schema:** Create a new database for the restored data.
 - **Existing Schema:** Overwrite an existing database (ensure no critical data is lost).
 6. **Click "Start Import"** and wait for the process to complete.
 7. **Verify the restored data** by querying tables in the SQL editor.
-

Lab Steps for Generating Schema

Step 1: Problem Statement Analysis

Scenario: Design a database for a **Library Management System** with:

- **Books** (ID, Title, Author, Published Year)
 - **Members** (ID, Name, Email, Join Date)
 - **Loans** (Loan ID, Book ID, Member ID, Due Date)
-

Step 2: Draw the ER Diagram

Use a tool like **draw.io** or **MySQL Workbench ERD**: ([Click here](#))

1. Goto File -> New Model and add followings:
 2. **Entities:** Represented as **rectangles**.
 - Book, Member, Loan
 3. **Attributes:** Listed as **ovals** connected to entities.
 - Book: book_id (PK), title, author
 4. **Relationships:** Shown as **diamonds** with cardinality.
 - Member → Loan (1:M)
 - Book → Loan (1:M)
-

Step 3: Convert ER Diagram to MySQL Schema

Create SQL script via:

- Goto Database -> Forward Engineer.
 - **Run the SQL Script:** Paste and execute the CREATE TABLE statements.
-

Step 5: Verify the Schema

1. Check Tables:

```
SHOW TABLES;
```

2. View Table Structure:

```
DESCRIBE Book;
```

```
DESCRIBE Loan;
```

Step 6: Test with Sample Data

Insert sample records and query relationships:

-- Insert a Member

```
INSERT INTO Member (name, email) VALUES ('Alice', 'alice@example.com');
```

-- Insert a Book

```
INSERT INTO Book (title, author) VALUES ('Database 101', 'John Doe');
```

-- Record a Loan

```
INSERT INTO Loan (member_id, book_id, due_date)
VALUES (1, 1, DATE_ADD(NOW(), INTERVAL 14 DAY));
```

-- Query Active Loans

```
SELECT m.name, b.title, l.due_date
FROM Loan l
JOIN Member m ON l.member_id = m.member_id
JOIN Book b ON l.book_id = b.book_id;
```

Exercises (Class)

Add here all the tasks performed in lab.

Exercises (Weekly)

1. Write step-by-step procedure for Backup and recovery of your database (with screenshot).
2. **Design an ER diagram** for a **University Database** (Students, Courses, Enrollments).
3. **Write SQL** to create the schema.
4. **Insert sample data** and run queries.